

Standalone assessment

Kubernetes Midterm Exam - Lec



Submissions

Here are your latest answers:

Question 1

5.The API Server acts as the central management entity that exposes the Kubernetes API.

Choices:

-

True

-

False

Response: True

Correct answer: True

Score: 1 out of 1 Yes

Question 2

13.A ReplicaSet ensures a specified number of pod replicas are running at all times.

Choices:

-

True

-

False

Response: True

Correct answer: True

Score: 1 out of 1 Yes

Question 3

12.The kubelet runs on every node and ensures containers are running as expected.

Choices:

-

True

-

False

Response: True

Correct answer: True

Score: 1 out of 1 Yes

Question 4

43.The Service type that provides access only within the cluster is called _____.

Response: ClusterIP

Correct answer: ClusterIP

Score: 1 out of 1 Yes

Question 5

35.Which object would you use to mount an external storage resource to a pod?

Choices:

-

Service

-

Persistent Volume Claim

-

ConfigMap

-

Secret

Response: Persistent Volume Claim

Correct answer: Persistent Volume Claim

Score: 1 out of 1 Yes

Question 6

3.Pods in Kubernetes can contain multiple containers that share the same network namespace.

Choices:

-

True

-

False

Response: True

Correct answer: True

Score: 1 out of 1 Yes

Question 7

8.The NodePort service type allows external access to services via a static port on each node.

Choices:

-

True

-

False

Response: True

Correct answer: True

Score: 1 out of 1 Yes

Question 8

45.The object that defines how Pods are created and updated in a controlled manner is called a _____.

Response: Deployment

Correct answer: Deployment

Score: 1 out of 1 Yes

Question 9

27.Which of the following best defines a ConfigMap?

Choices:

-

A method to encrypt secret data

-

A mechanism to store non-confidential configuration data

-

A network routing service

-

A persistent storage interface

Response: A mechanism to store non-confidential configuration data

Correct answer: A mechanism to store non-confidential configuration data

Score: 1 out of 1 Yes

Question 10

30.What is the purpose of a Deployment in Kubernetes?

Choices:

- To manage database connections
- To automate updates and scaling of pods
- To provide monitoring dashboards
- To manage networking policies

Response: To automate updates and scaling of pods

Correct answer: To automate updates and scaling of pods

Score: 1 out of 1 Yes

Question 11

36.The main configuration file used to define Kubernetes objects is called a _____ file.

Response: YAML

Correct answer: YAML

Score: 1 out of 1 Yes

Question 12

22.Which Service type exposes a service externally using a cloud provider's load balancer?

Choices:

- ClusterIP
- NodePort
- LoadBalancer
- ExternalName

Response: LoadBalancer

Correct answer: LoadBalancer

Score: 1 out of 1 Yes

Question 13

15.Kubernetes uses YAML files to define objects declaratively.

Choices:

- True

-
False

Response: True

Correct answer: True

Score: 1 out of 1 Yes

Question 14

28. Persistent Volume Claims (PVC) are used by:

Choices:

- Pods to request storage resources
- Nodes to communicate with the API server
- Services to manage traffic
- The scheduler to allocate memory

Response: Pods to request storage resources

Correct answer: Pods to request storage resources

Score: 1 out of 1 Yes

Question 15

37. The Kubernetes command-line interface used to communicate with the cluster is called ____.

Response: kubectl

Correct answer: kubectl

Score: 1 out of 1 Yes

Question 16

41. In Kubernetes networking, every Pod has its own unique ____ address.

Response: IP

Correct answer: IP

Score: 1 out of 1 Yes

Question 17

49. A Pod can request specific storage space by creating a ____.

Response: Persistent Volume Claim (PVC)

Correct answer: Persistent Volume Claim (PVC)

Score: 1 out of 1 Yes

Question 18

18.Which component in the control plane maintains cluster state and configuration data?

Choices:

- Scheduler
- etcd
- kubelet
- Controller Manager

Response: etcd

Correct answer: etcd

Score: 1 out of 1 Yes

Question 19

14.LoadBalancer services are typically used only in on-premise Kubernetes environments.

Choices:

- True
- False

Response: False

Correct answer: False

Score: 1 out of 1 Yes

Question 20

38.The smallest deployable computing unit in Kubernetes is the _____.

Response: Pod

Correct answer: Pod

Score: 1 out of 1 Yes

Question 21

47.The object used to store and manage sensitive data such as passwords or API tokens is a _____.

Response: Secret

Correct answer: Secret

Score: 1 out of 1 Yes

Question 22

21.What does a Kubernetes Service provide?

Choices:

- Container build automation
- Persistent storage
- A stable network endpoint for pods
- Monitoring of pod logs

Response: A stable network endpoint for pods

Correct answer: A stable network endpoint for pods

Score: 1 out of 1 Yes

Question 23

23.Which file format is commonly used for defining Kubernetes manifests?

Choices:

- JSON
- YAML
- XML
- INI

Response: YAML

Correct answer: YAML

Score: 1 out of 1 Yes

Question 24

4.The kubectl command-line tool is used to interact with the Kubernetes cluster.

Choices:

- True
- False

Response: True

Correct answer: True

Score: 1 out of 1 Yes

Question 25

44.The Service type that provisions an external IP address using a cloud provider is called _____.

Response: LoadBalancer

Correct answer: LoadBalancer

Score: 1 out of 1 Yes

Question 26

29.Which of the following is not a Kubernetes Service type?

Choices:

- ClusterIP
- NodePort
- LoadBalancer
- ReplicaPort

Response: ReplicaPort

Correct answer: ReplicaPort

Score: 1 out of 1 Yes

Question 27

25.A Namespace in Kubernetes is used to:

Choices:

- Manage networking policies
- Organize cluster resources logically
- Store container images
-

Scale pods horizontally

Response: Organize cluster resources logically

Correct answer: Organize cluster resources logically

Score: 1 out of 1 Yes

Question 28

42.The Service type that exposes an application on a specific port of each node is called _____.

Response: NodePort

Correct answer: NodePort

Score: 1 out of 1 Yes

Question 29

50.Kubernetes automatically restarts failed Pods through the _____ controller.

Response: ReplicaSet

Correct answer: ReplicaSet

Score: 1 out of 1 Yes

Question 30

39.A group of one or more containers that share storage and network resources is called a _____.

Response: Pod

Correct answer: Pod

Score: 1 out of 1 Yes

Question 31

17.What is the smallest deployable unit in Kubernetes?

Choices:

- Node
- Pod
- Service
- Deployment

Response: Pod

Correct answer: Pod

Score: 1 out of 1 Yes

Question 32

34.The term desired state in Kubernetes refers to:

Choices:

- The system's default configuration
- The user-defined configuration that Kubernetes tries to maintain
- The temporary runtime state
- The physical machine configuration

Response: The user-defined configuration that Kubernetes tries to maintain

Correct answer: The user-defined configuration that Kubernetes tries to maintain

Score: 1 out of 1 Yes

Question 33

16.Which of the following best describes Kubernetes?

Choices:

- A database system
- A container orchestration platform
- A programming framework
- A virtual machine manager

Response: A container orchestration platform

Correct answer: A container orchestration platform

Score: 1 out of 1 Yes

Question 34

10.Persistent Volumes (PV) and Persistent Volume Claims (PVC) are used for managing storage in Kubernetes.

Choices:

- True
- False

Response: True

Correct answer: True

Score: 1 out of 1 Yes

Question 35

31.Which command would you use to scale a deployment named “web-app” to 5 replicas?

Choices:

- kubectl scale web-app=5
- kubectl scale deployment web-app --replicas=5
- kubectl replicas web-app 5
- kubectl update web-app --replicas 5

Response: kubectl scale deployment web-app --replicas=5

Correct answer: kubectl scale deployment web-app --replicas=5

Score: 1 out of 1 Yes

Question 36

26.The kube-proxy handles:

Choices:

- Storage management
- Network routing and load balancing
- Pod scaling
- Security configurations

Response: Network routing and load balancing

Correct answer: Network routing and load balancing

Score: 1 out of 1 Yes

Question 37

9.A Kubernetes ConfigMap is used to store sensitive data such as passwords.

Choices:

- True

-
False

Response: False

Correct answer: False

Score: 1 out of 1 Yes

Question 38

2.A Kubernetes cluster consists of one or more master nodes only.

Choices:

-
True
-
False

Response: False

Correct answer: False

Score: 1 out of 1 Yes

Question 39

20.Which command is used to view all running pods in a namespace?

Choices:

-
kubectl pods get all
-
kubectl get pods
-
kubectl describe pods
-
kubectl get services

Response: kubectl get pods

Correct answer: kubectl get pods

Score: 1 out of 1 Yes

Question 40

19.Which Kubernetes component is responsible for scheduling Pods to Nodes?

Choices:

-
API Server
-

Scheduler

-

etcd

-

kube-proxy

Response: Scheduler

Correct answer: Scheduler

Score: 1 out of 1 Yes

Question 41

48.The storage resource that provides long-term data persistence independent of Pods is a _____.

Response: Persistent Volume

Correct answer: Persistent Volume

Score: 1 out of 1 Yes

Question 42

40.The process of ensuring the actual state of the cluster matches the desired state is known as _____.

Response: Reconciliation

Correct answer: Reconciliation

Score: 1 out of 1 Yes

Question 43

1.Kubernetes is an open-source platform designed to automate deploying, scaling, and operating containerized applications.

Choices:

-

True

-

False

Response: True

Correct answer: True

Score: 1 out of 1 Yes

Question 44

6.The Scheduler is responsible for ensuring containers are running and healthy.

Choices:

-

True

-

False

Response: False

Correct answer: False

Score: 1 out of 1 Yes

Question 45

32.A Pod is scheduled to run on which component?

Choices:

-

API Server

-

Node

-

etcd

-

Controller

Response: Node

Correct answer: Node

Score: 1 out of 1 Yes

Question 46

7.ClusterIP is the default service type in Kubernetes that exposes services only inside the cluster.

Choices:

-

True

-

False

Response: True

Correct answer: True

Score: 1 out of 1 Yes

Question 47

24.The command `kubectl apply -f deployment.yaml` is used to:

Choices:

- Delete a deployment
- Apply a configuration file to the cluster
- Check cluster nodes
- Restart all pods

Response: Apply a configuration file to the cluster

Correct answer: Apply a configuration file to the cluster

Score: 1 out of 1 Yes

Question 48

46.The Kubernetes object used to inject environment-specific data into a container is called a _____.

Response: ConfigMap

Correct answer: ConfigMap

Score: 1 out of 1 Yes

Question 49

33.What does the term container orchestration mean?

Choices:

- Creating containers
- Managing the lifecycle and scaling of containers
- Monitoring databases
- Writing Dockerfiles

Response: Managing the lifecycle and scaling of containers

Correct answer: Managing the lifecycle and scaling of containers

Score: 1 out of 1 Yes

Question 50

11.The Controller Manager manages various controllers that regulate the cluster's state.

Choices:

- True
-

False

Response: True

Correct answer: True

Score: 1 out of 1 Yes